



Seismic radial anisotropy in southeastern Tibetan Plateau and its implications for regional geodynamic evolution

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ABSTRACT

The southeastern Tibetan Plateau exhibits intricate crustal tectonics, encompassing recent seismic megathrust events. Previous research suggested the presence of north-south-oriented channelized viscous flow within the crust. However, recent investigations have unveiled a notable northeast-southwest-oriented geological structure, potentially rigid, intersecting with the presumed crustal channelized flow. Several questions persist regarding the composition of the northeast-southwest-oriented structure, the continuity of crustal channelized flow, and the interplay between them. In this study, dispersion data from a dense seismic array are employed to significantly refine regional crustal models for shear wave velocity and radial anisotropy through ambient noise tomography. The resulting high-resolution model further reveals the style of the crustal deformation and supports the interpretation that the northeast-southwest structure, which shows higher velocity and significant negative radial anisotropy, results from mafic material at the base of crust, obstructing the crustal channelized flow. However, the northeast-southwest structure is not as rigid as the Sichuan Block and exhibits depth-dependent deformation. The interpretation proves useful in further understanding regional earthquake focal mechanisms and strain distribution. Additionally, this research identified a region of generalized negative radial anisotropy in the crust of the western Chuan-Dian fragment, suggesting a reduced horizontal channel crustal flow in this area. Drawing upon various geophysical and geological evidence, we present a geodynamic evolution model, proposing a sequence of events: Permian plume activity resulting in mafic material at the base of the crust near Anninghe-Zemuhe fault, northward advancement of the east Himalayan syntaxis inducing crustal compressional stress field, reduced lower crustal channel flow in the western Chuan-Dian fragment leading to the regional east-west extension, and initiation of the Xianshuihe fault causing shift of strain concentration and depth-dependent deformation near the Anninghe-Zemuhe fault. The geodynamic model provides valuable insights into the regional distribution of crustal strain and the underlying mechanisms of large seismic events.

1. Introduction

The southeastern Tibetan Plateau (SETP; Fig. 1) has been a focal point of extensive geological and geophysical investigations because of the quest to understand the eastward expansion of the plateau and its role in seismic activity (Liu et al., 2014). The latter is manifest by a series of significant recent seismic events, including the Wenchuan 2008

Mw 7.9 earthquake, Lushan 2013 Mw 6.6 earthquake, Lijiang 1996 Mw 6.6 earthquake, and the recent Luding 2022 Mw 6.7 earthquake. These events, denoted in Fig. 1c (marked in red, green, and blue beach balls, respectively), have spurred interest in comprehending the crustal structures of this region (Zhang, 2013; Yan et al., 2018). The geology of SETP is further compounded by the presence of world-class ore deposits (Pang et al., 2010; Munteanu et al., 2013), particularly along the Panxi re-

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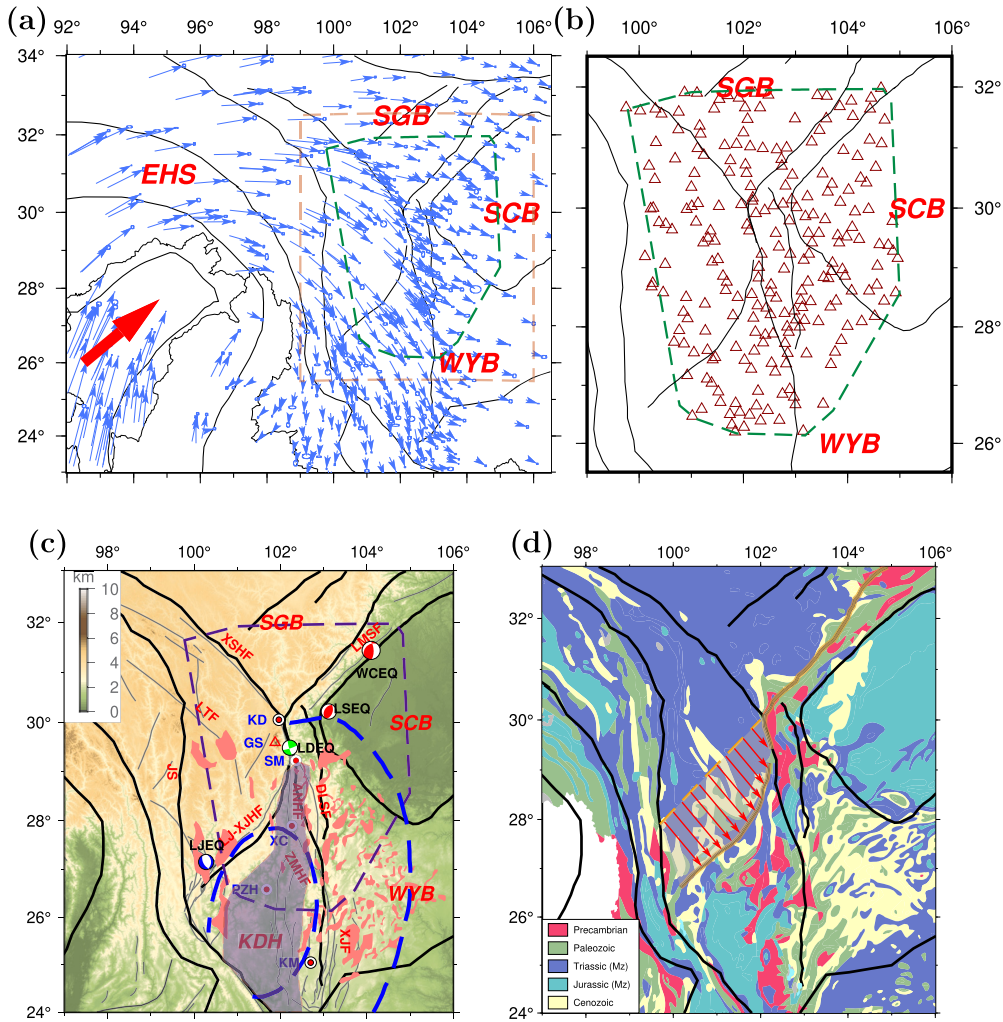


Fig. 1. GPS observation, seismic station distribution, topography and geological maps. (a) GPS observation map. The blue arrows are the GPS observations with respect to stable Eurasia collected by Zheng et al. (2017). The red arrow is the movement direction of the east Himalayan syntaxis. The red dashed line outlines the study region in (b). (b) Seismic station distribution. The red triangles denote the detailed distribution of the stations. (c) Topography map. The blue dashed lines outline the boundaries of the inner and intermediate regions of the Emeishan Large Igneous Province. The red, green, and blue beach balls indicate the large earthquakes with reverse-, strike-slip, and normal-fault motion, respectively. (d) Geological map. The solid brown line denotes the Longmenshan Fault and Lijiang-Xiaojinhe Fault, which generally separate the Triassic and Jurassic exposures. The dashed yellow line is inferred initial location of the Lijiang-Xiaojinhe fault recovered by extrapolation of GPS observations (Gan et al., 2022). The red arrows represent the movement direction of the fault. The geological data are from Steinshouer et al. (1999). Abbreviations: SGB, Songpan-Ganze Block; SCB, Sichuan Basin; WYB, Western Yangtze Block; KDH, Kungdian High; EHS, the eastern Himalayan syntaxis; LMSF, Longmenshan Fault; XSHF, Xianshuihe Fault; LJ-XJHF, Lijiang-Xiaojinhe Fault; ANHF, Anninghe Fault; DLSF, Daliangshan Fault; ZMHF, Zemuhe Fault; XJF, Xiaojiang Fault; LTF, Litang Fault; JS, Jinsha Suture; KD, Kangding; GS, Gonggashan; SM, Shimian; XC, Xichang; PZH, Panzhihua; KM, Kunming; WCEQ, Wenchuan earthquake; LSEQ, Lushan earthquake; LJEQ, Lijiang earthquake; LDEQ, Luding earthquake.

gion, a narrow north-south-oriented zone situated between the cities of Panzhihua (PZH) and Xichang (XC) in western Sichuan Province (see Fig. 1c).

Crustal channel flow (Royden et al., 1997, 2008), albeit with substantial lateral heterogeneity (e.g., Liu et al., 2014), has been proposed as a framework to comprehend the intricate tectonic structure and diverse geophysical observations in SETP. Details are still debated, but pervasive low-velocity (and presumably mechanically weak) anomalies ($V_S < 3.4$ km/s) in the mid-lower crust of SETP are evident in various velocity models (Yao et al., 2008; Li et al., 2009; Liu et al., 2014; Bao et al., 2015; Fu et al., 2017; Yang et al., 2020b; Zhang et al., 2020; Yang et al., 2020a; Liu et al., 2023). Additionally, magnetotelluric studies have identified two corresponding N-S directional low-resistivity belts (Bai et al., 2010). These observations, combined with GPS and SKS studies (Wu et al., 2015; Liu et al., 2020), lend credence to the presence of crustal channel flow model in this region. Initially introduced to explain surficial topography variations with relatively mild Moho variations (Royden

et al., 1997), the model posits that the Cenozoic collision (Yin and Harrison, 2000) between the Eurasian and Indian plates results in crustal thickening, stress depression, and partial melt in the mid-lower crust. Subsequently, material in the mid-lower crust decouples from the mantle and the upper crust, migrates from central Tibetan Plateau to the eastern Himalayan syntaxis, and proceeds southeastward (Royden et al., 2008; Hu and Yao, 2018). Recent studies propose the existence of two crustal channel flows (Bai et al., 2010; Bao et al., 2015) bifurcated around the Panzhihua area (PZH; Fig. 1c). The presence and geometry of a weak crustal zone can explain high seismicity along the Longmenshan fault (LMSF) and the association with the Wenchuan 2008 Mw 7.9 earthquake (Burchfiel et al., 2008; Wang et al., 2011), and the Middle-Late Miocene (13 – 8 Ma) tectonic transformation in the upper crust (Zhang et al., 2023a).

Furthermore, the region hosts world-class ore deposits, particularly along the Panxi region, a N-S directional narrow zone between Panzhihua (PZH) and Xichang (XC) in western Sichuan Province (Fig. 1c).

Panxi is centered at the Emeishan Large Igneous Province (ELIP) (Xu et al., 2006; Chen et al., 2015), and seismic studies reveal a crustal structure that is around Panzhihua characterized by high velocity (Chen et al., 2015), Moho elevation, and a gravity anomaly (Deng et al., 2014). The formation of the Panxi region is possibly related to ancient mantle plume activity around 260 Ma (Xu et al., 2006). The relatively rigid crust in this region is perceived as a barrier to southward channel flow (Yang et al., 2020b; Zhang et al., 2020; Li et al., 2020). Despite its successful use in explaining geophysical observations, the validity of the “two crustal channel flow” model has been questioned by recent studies (Chen et al., 2014; Zhang et al., 2020; Bao et al., 2020). High-resolution seismic velocity models indicate that low velocities are discontinuous from the northwest and southeast (Chen et al., 2014; Yang et al., 2020b; Zhang et al., 2020; Han et al., 2022), and azimuthal anisotropy also suggests a more complex pattern of channelized flow (Bao et al., 2020; Cao et al., 2023). If the proposed channel flow is obstructed by a zone of high seismic velocities (suggesting high mechanical strength), strain heterogeneity in this region may affect seismicity and seismic anisotropy.

A particular area of interest is seismic radial anisotropy, which differentiates between the propagation speeds of vertically and horizontally polarized waves, offering insights into material alignment and strain conditions (Shapiro et al., 2004; Huang et al., 2010; Xie et al., 2013; Hu et al., 2020). A zone displaying negative radial anisotropy in the NE-SW direction was identified in western Sichuan (Xie et al., 2013, 2017; Liu et al., 2021). However, its origins remain subject to debate, ranging from mineral alignment (Xie et al., 2013, 2017) to a concealed hotspot track (Liu et al., 2021; Hu et al., 2023). Given that this zone intersects one of the proposed pathways for “channel flow” (Bai et al., 2010) and is positioned within the inner and intermediate zones of the ELIP, it is imperative to understand how this radial anisotropy structure aligns with the channel flow model and its potential relationship with plume-related magmatism.

Over recent years, a substantial body of seismic research has harnessed the data from the dense seismic array in this region. These studies encompass a range of methodologies, including joint inversion for shear wave velocity (Liu et al., 2014; Peng et al., 2023), teleseismic receiver function analysis (Wang et al., 2017; Xu et al., 2020; Hu et al., 2023), azimuthal anisotropy investigations (Zheng et al., 2018; Cao et al., 2020; Liu et al., 2020; Han et al., 2022; Cao et al., 2023), attenuation tomography (Liu et al., 2024), and analyses of surficial stress distribution (Yang et al., 2018). Collectively, these investigations have significantly enhanced our comprehension of regional tectonics.

The study presented here leverages continuous seismic data from a dense seismic array in western Sichuan (Liu et al., 2014) to refine the regional shear wave velocity seismic radial anisotropy model. The research aims to establish the link between radial anisotropy and the spatial distribution of regional seismicity, as well as its association with earthquake focal mechanisms. Ultimately, the study integrates various geophysical observations to propose a comprehensive regional geodynamic evolution model, shedding light on regional strain distribution and the causative factors behind significant earthquakes. The complexity and significance of the geological and tectonic attributes of SETP underscore the importance of this research in advancing our understanding of this dynamic region and its seismic hazards.

2. Geological and tectonic context

Fig. 1d provides important insights into the geological history of the region, particularly the separation of exposures from Triassic and Jurassic times by two significant fault systems: the LMSF and the Lijiang-Xiaojinhe fault (LJ-XJHF) in the NE-SW direction (Solid brown line in Fig. 1d). Geological evidence suggests significant shortening of the adjacent Songpan-Ganze Block (SGB) and Yangtze Block (YB) during the Late Triassic-Early Jurassic period (Burchfiel et al., 1995; Burchfiel and Chen, 2012).

Furthermore, previous geological investigations have established the LJ-XJHF as a southern extension of the LMSF, formed during both the Mesozoic and Cenozoic eras (since 250 Ma) (Burchfiel and Chen, 2012). Notably, a recent study by Gan et al. (2022) indicates that the LMSF and LJ-XJHF align almost along a straight line when their current GPS rates are extrapolated backward in time since ~10 Ma (Dashed yellow line in Fig. 1d). The differences in focal mechanisms observed for major earthquakes along the LMSF and LJ-XJHF (Fig. 1c) suggest distinct compressional and extensional tectonic regimes operating along these faults.

Additionally, Precambrian rock exposures are found within the north-south-oriented Kungdian High region (KDH, shaded in purple in Fig. 1c). This paleogeographic element has a history of uplift during the Paleozoic and early Mesozoic periods (Burchfiel and Chen, 2012). The ore-enriched Panxi region, situated within the KDH, is also located within the inner and intermediate regions of the ELIP (indicated by the dashed line in Fig. 1c).

Another noteworthy feature is the geological age of rocks on the western Yangtze Block (WYB), east of the Xiaojiang fault (XJF), which primarily dates back to the Cenozoic era (~66 Ma). In contrast, the Sichuan Block (SCB) and the region near Panzhihua (PZH) city in the west of XJF exhibit relatively older and stable geological characteristics. Several studies have estimated the initiation ages of exhumation of major mountains and the commencement of major faults. The Xianshuihe fault (XSHF) and XJF initiation ages are estimated to be around 10 Ma (Zhang, 2013). Rapid uplift and exhumation of Gonggashan mountain (GS) are traced back to around 9 Ma (Zhang et al., 2019), while the initiation of LMSF is estimated to have occurred around 13 Ma (Shen et al., 2019). Gan et al. (2022) also suggest that the deformation of LMSF and LJ-XJHF began around ~10 Ma. Given the proximity in time between the initiation of the XSHF and XJF and the rapid uplift and exhumation of GS and LMSF, it is plausible to assume that these tectonic events are interrelated or share a common driving force (Zhang et al., 2023a).

An additional point of significance is the clockwise rotation of the SETP around the eastern Himalayan syntaxis (EHS) as revealed by GPS data (Fig. 1a). Previous studies have traced the onset of this rotation to the late Oligocene (~25.7 Ma), likely induced by the indentation of the Indian plate since ~28 Ma (Tong et al., 2021).

Understanding the intricate interplay of these geological and tectonic processes is pivotal to unraveling the dynamic evolution and seismic hazards in this region.

3. Seismic data collection and analysis

Continuous seismic data were collected from a densely deployed seismic array in western Sichuan (see Fig. 1b). This seismic array was operational from 2006 to 2009 and featured an average interstation spacing of 15 km (Liu et al., 2014).

In this study, we harnessed the continuous vertical-vertical and transverse-transverse components of the seismic data to extract empirical Green's functions (Figures S2a and S2b). In this context, ‘vertical-vertical’ refers to the cross-correlation of the vertical components between two stations, while ‘transverse-transverse’ refers to the cross-correlation of the transverse components between two stations. Dispersion data were then derived through ambient noise interferometry and the image transformation method (Yao et al., 2006; Huang et al., 2010). Both Rayleigh and Love wave dispersion was extracted for periods ranging from 4 to 40 seconds. As a quality control measure, only the empirical Green's functions with signal-to-noise ratio larger than 10 were used for further analysis (Huang, 2014). Another criterion when extracting the dispersion curves is that the distance between stations needed to exceed twice the wavelength.

As a result, there are totally 11,987 inter-station Green's functions for Rayleigh wave and 6,693 for Love wave, respectively. Our dataset comprises a total of 226,220 Rayleigh wave arrival times and 90,491

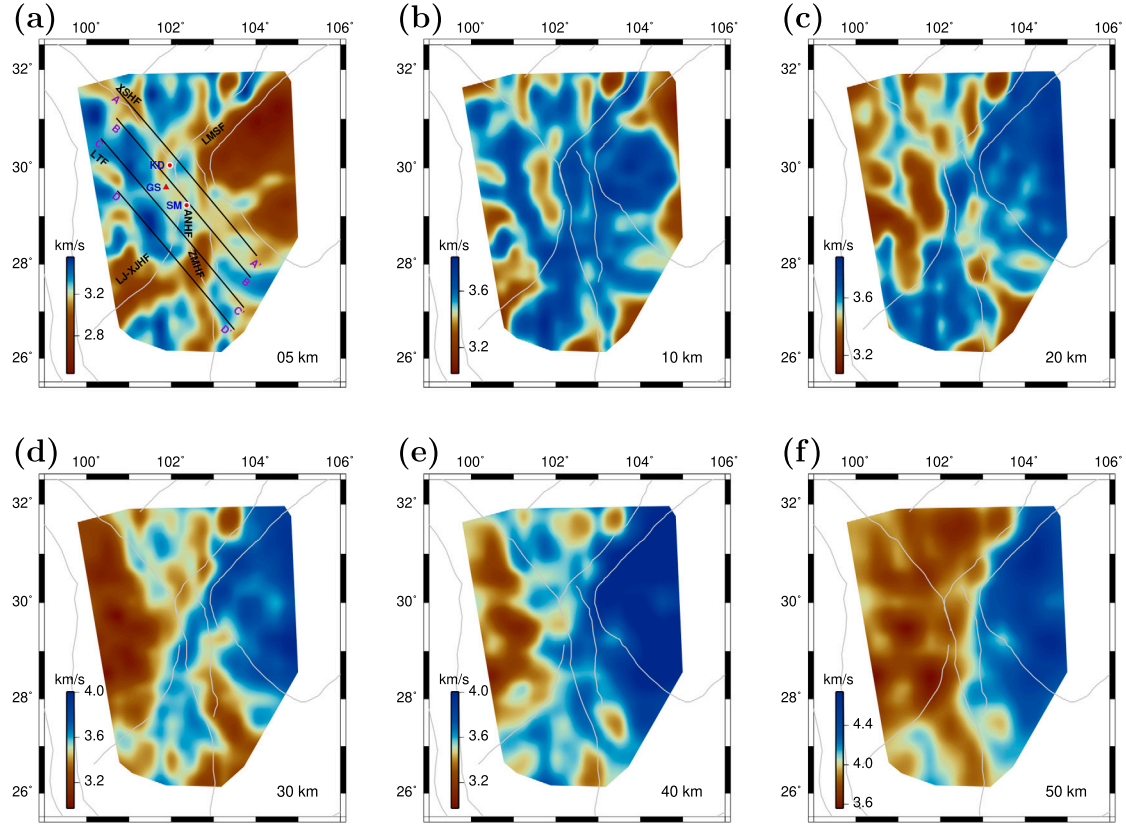


Fig. 2. Average shear wave velocity V_s at different depths. Major faults and major cities are labeled in (a). The four straight lines in (a) denote the locations of the cross sections in Fig. 4. Abbreviations: LMSF, Longmenshan Fault; XSHF, Xianshuihe Fault; LJ-XJHF, Lijiang-Xiaojinhe Fault; ANHF, Anninghe Fault; ZMHF, Zemuhe Fault; LTF, Litang Fault; KD, Kangding; GS, Gonggashan; SM, Shimian.

Love wave arrival times (see Figures S2c and S2d). For detailed information regarding data processing, please refer to Huang (2014).

Using the dispersion curves thus obtained, we conducted seismic radial anisotropy inversion employing the direct radial anisotropy tomography method (DRadiSurfTomo) (Hu et al., 2020; Yang et al., 2020a). A brief description of the method can be found in the supplementary material. This method accounts for ray-bending effects and enables simultaneous inversion of V_{sv} and $\frac{V_{sh}}{V_{sv}}$. Then the radial anisotropy (ξ) and average shear wave velocity (V_s) can be calculated. In this study, the radial anisotropy ξ is defined as $\frac{2(V_{sh}-V_{sv})}{V_{sv}+V_{sh}}$ and the average shear wave velocity V_s is defined as $\frac{V_{sv}+V_{sh}}{2}$. In positive radial anisotropy, the horizontal shear wave velocity V_{sh} is faster than the vertical shear wave velocity V_{sv} , while negative radial anisotropy is indicated by V_{sv} being faster than V_{sh} . The smoothing parameters were chosen by the L-curve analysis (see Figure S7 for details). Checkerboard tests suggest that structure patterns with lateral dimensions larger than $0.5^\circ \times 0.5^\circ$ can be resolved (see Figures S5 and S6).

To assess the reliability of our results, we employed the jackknife resampling technique (Hu et al., 2020). This resampling technique allowed us to estimate the standard deviation of the inversion. The results of the jackknife resampling demonstrated that the standard deviations were generally around 4% for radial anisotropy (ξ) and smaller than 0.15 km/s for shear wave velocity (V_s) across most of the study region (see Figures S9 and S10). The recovery tests demonstrate that the main features of the anisotropy model are stable, and the errors introduced in the recovered V_s model appear to be acceptable (Figures S11 and S12).

4. Results

Fig. 2 reveals distinct features in the V_s model, which closely correlate with the tectonic blocks within the study region. Firstly, within the

SCB, there is a low-velocity anomaly at a depth of approximately 5 km, juxtaposed to a high-velocity region at depths ranging from 10 to 50 km. Secondly, the Songpan-Ganze Block (SGB) displays a consistent pattern of low velocity at depths below 20 km, likely indicating a mechanically weak mid-lower crust. The mechanically weak mid-lower crust characterized by low velocity has been widely recognized within the Tibetan Plateau (Yang et al., 2012). Moreover, an intriguing high-velocity anomaly is observed near the Anninghe fault (ANHF) at depths of 20–40 km, intersecting the low-velocity zones in the northwest and southeast of the study area. The high-velocity anomaly in the mid-crust, possibly indicative of a relatively rigid area, may influence regional stress accumulation and seismic distribution. Additionally, to the west of Kangding city (KD) and Shimian city (SM) at a depth of 10 km, a clear north-south directional pattern emerges, outlining the GS mountain.

Fig. 3 illustrates the radial anisotropy patterns across various depths and tectonic features, some of which are of particular interest and will be discussed in more detail below. Firstly, At shallow depths of 5 km, bands of negative radial anisotropy ($V_{sh} < V_{sv}$) are predominantly observed, aligned with the locations of major fault such as the XSHF, LMSF, LJ-XJHF, and XJF. Mirroring the V_s map, the GS mountain demonstrates distinct north-south trending features of positive radial anisotropy, also at a depth of 5 km. Secondly, at the depth of 10 km, there is apparent positive radial anisotropy around the ZMHF. By analyzing soil gas CO_2 data, Yang et al. (2021) revealed that the ZMHF has less level of coupling, including the locking depth and extent, compared to the ANHF. Since the radial anisotropy in the upper crust is likely to be caused by the alignment of stress- or structure-induced fluid-saturated cracks (Acevedo et al., 2022), the positive radial anisotropy at 10 km around the ZMHF in our model probably results from the horizontally aligned fluid-saturated cracks of the unhealed fault. Thirdly, at depths below 20 km, a prominent feature is the presence of positive radial anisotropy zone in the SGB, which is also characterized by low veloc-

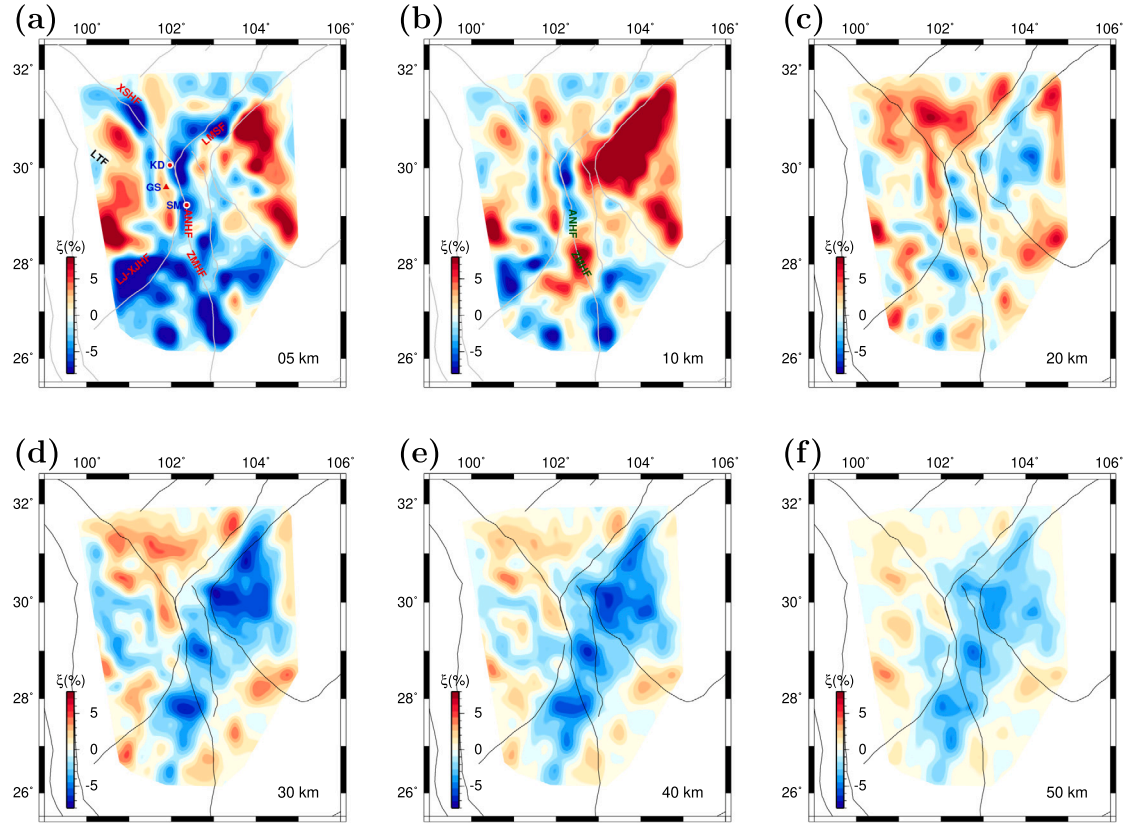


Fig. 3. Radial anisotropy ξ at different depths. Major faults and major cities are labeled in (a). The definitions of abbreviations are the same as those in Fig. 2.

ity. The agreement of the low velocity and positive radial anisotropy possibly indicates the mid-lower crustal material is mechanically weak and that the minerals align in the horizontal direction. The mechanically weak crust with horizontally aligned minerals was thought to play a crucial role in forming the broad and flat Tibetan Plateau (Royden et al., 1997, 2008). Fourthly, a NE-SW trending of negative radial anisotropy zone east of but roughly parallel to the LMSF and LJ-XJHF emerges at 20 km depth and is prominent in the 30–50 km depth range. The anisotropy structure coincides spatially with the NE-SW trending zone of high seismic velocities at similar depths (30–40 km) beneath the western part of the Sichuan Basin. Further examination reveals that the negative radial anisotropy feature can be subdivided into three sub-regions, separated by the north-south oriented Daliangshan fault (DLSF) and the Anninghe-Zemuhe fault (ANH-ZMHF). Liu et al. (2021) indicated that the characterization of the magma storage system as having negative or positive radial anisotropy largely depends on whether it is dike-dominated (like the ELIP) or sill-dominated. In our model, the observed negative radial anisotropy and high velocity may suggest the presence of a dike-dominated system, where the crustal material is relatively rigid with minerals aligned in the vertical direction. Note the intrusion of ELIP does not exhibit circular shape probably due to the dramatic tectonic evolution since ELIP eruption around 260 Ma, such as the formation of the LJ-XJHF and the movement of the mid-lower crustal material. It is also possible to attribute the NE-SW directional distribution of intrusion to the hidden hotspot track (Liu et al., 2021). Another substantial negative radial anisotropy zone is evident in the western region of the XSHF and to the north of the LJ-XJHF, a region often referred to as the western Chuan-Dian fragment. This negative radial anisotropy is prominent at depths from 30 to 40 km. The negative radial anisotropy seems to contradict the interpretation of existence of crustal channelized flow in this region.

In the vertical cross-sections of the shear wave velocity model (Figs. 4a, c, e, g), distinct low-velocity zones are observed between 20 and 40 km in the northwest. Additionally, there is an offset in the high-

velocity zones that spans depths of 10 to 40 km, intersecting both the LMSF and LJ-XJHF.

In the corresponding cross-sections of the radial anisotropy model (Figs. 4b, d, f, h), a general pattern emerges where positive radial anisotropy aligns spatially with the low-velocity zones in the mid-lower crust of the northwest. Conversely, negative radial anisotropy typically corresponds to areas of high velocity. The latter correlation is most prominent beneath the major fault zones, east of which a layer of positive radial anisotropy is visible between 10 and 20 km. It is also interesting to note that the low-reliefs (near offset 0 km) are correlated well to the high-velocity and negative radial anisotropy in the mid-lower crust. A possible explanation of the spatial correlation of low-reliefs, high velocity, and negative radial anisotropy is that the rigid mid-lower crust (characterized by high velocity) interacts with the horizontal movement of the mechanically weak flow (characterized by low velocity), thereby affecting the surficial topography.

5. Discussion

5.1. Constraints on crustal channel flow

Previous high-resolution velocity models include the joint inversion of surface wave dispersion and teleseismic body wave by Liu et al. (2014), Yang et al. (2020b), Peng et al. (2023) and Yang et al. (2023), ambient noise waveform tomography by Chen et al. (2014), and the community velocity model of southwest China developed by Liu et al. (2023). It is worth mentioning that most previous models, except Huang et al. (2010), primarily focus on V_{sv} , while our model includes V_s , a combination of V_{sv} and V_{sh} , offering a more comprehensive representation of the underlying rock properties.

Along with these studies, the results presented provide insight that can help constrain the existence, geometry, and continuity of crustal flow. First, our results confirm the presence of a pronounced negative radial anisotropy structure trending NE-SW along the major boundary

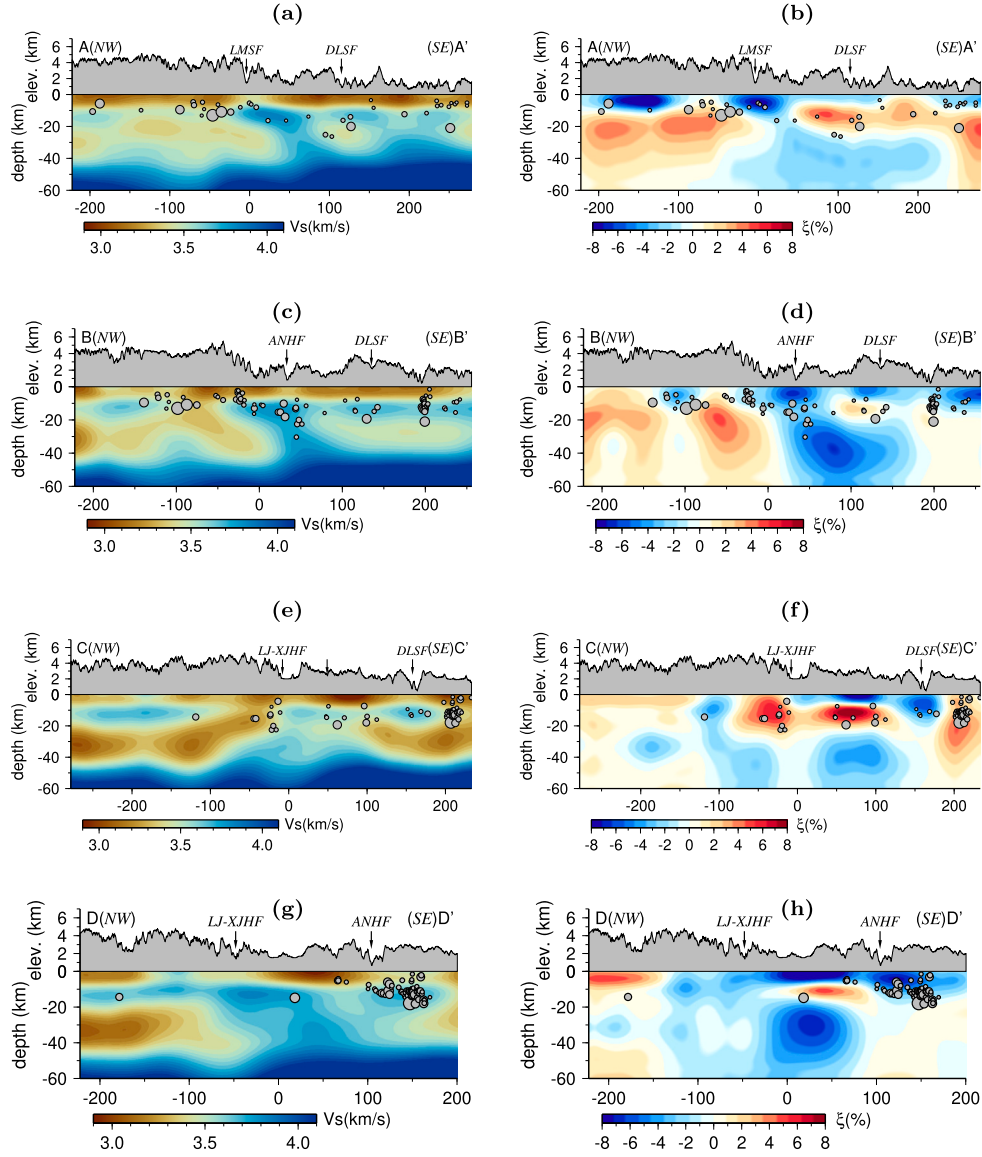


Fig. 4. Cross sections of the shear wave velocities (a, c, e, g) and radial anisotropy (b, d, f, h). The locations of the cross sections are shown in Fig. 2a. The gray circles denote the locations of the relocated earthquakes with magnitude larger than 1.5 during 2009/01–2019/03. The earthquake catalog is obtained from <http://www.cses.ac.cn/sjcp/ggmex>.

faults (LMSF and LJ-XJHF), that mark the easternmost boundary of the plateau (Xie et al., 2017; Liu et al., 2021). The interpretation of the boundary of the plateau is also supported by observed NE-SW directional azimuthal anisotropy along the LJ-XJHF and LMSF in the crust and upper mantle (Zhang et al., 2023b). In contrast to earlier studies, however, our images suggest that this large-scale structure is intersected by, and thus, possibly older than, N-S trending fault zone (notably, the DLSF and the ANH-ZMHF).

Second, similar to previous studies, our images reveal a high-velocity zone that is roughly parallel to the NE-SW oriented feature of negative radial anisotropy. This high velocity zone appears to cut through continuous low-velocity regions, extending from the SCB in the northeast to Panzhihua City in the southwest. Independent attenuation tomography has suggested that the material within this high-velocity structure is relatively rigid (He et al., 2021). Furthermore, using RF common-conversion-point stacking, Xu et al. (2020) identified a sharp lateral Moho variation in the NE-SW direction, approximately within the same region. Combining P wave and S wave RFs (Hu et al., 2023), as well as joint inversion RF waveforms and dispersion curves (Peng et al., 2023), Hu et al. (2023) and Peng et al. (2023) found a high velocity zone at

base of the crust, which they interpreted as the mafic underplated layer due to the ELIP activity.

Combining the constraints from RFs on crustal structure and from radial anisotropy on crustal fabric, we interpret the NE-SW directional belt-zone with fast wave propagation and negative radial anisotropy to be due to the existence of voluminous mafic intrusions and underplating. The high velocity of the NE-SW belt zone may then result from the mafic composition of the lower crust. This interpretation is further supported by the relatively high V_p/V_s ratio in this region (Wang et al., 2017). Together, the inference of low mid-crustal velocities (and low mechanical strength) below the eastern part of the plateau and a NE-SW structure of fast wave propagation, high mechanical strength, and near-vertical fabric further east suggest that lateral crustal flow from the central Tibetan Plateau (if any) is obstructed by a NE-SW trending structure that is related to intrusion of mafic materials (Chen et al., 2014; Yang et al., 2020b; Zhang et al., 2020; Liu et al., 2023; Han et al., 2022).

We note, however, that the NE-SW belt zone is not as strong as the SCB, and that the deformation pattern within this area may still be depth-dependent, as suggested by seismic azimuthal anisotropy stud-

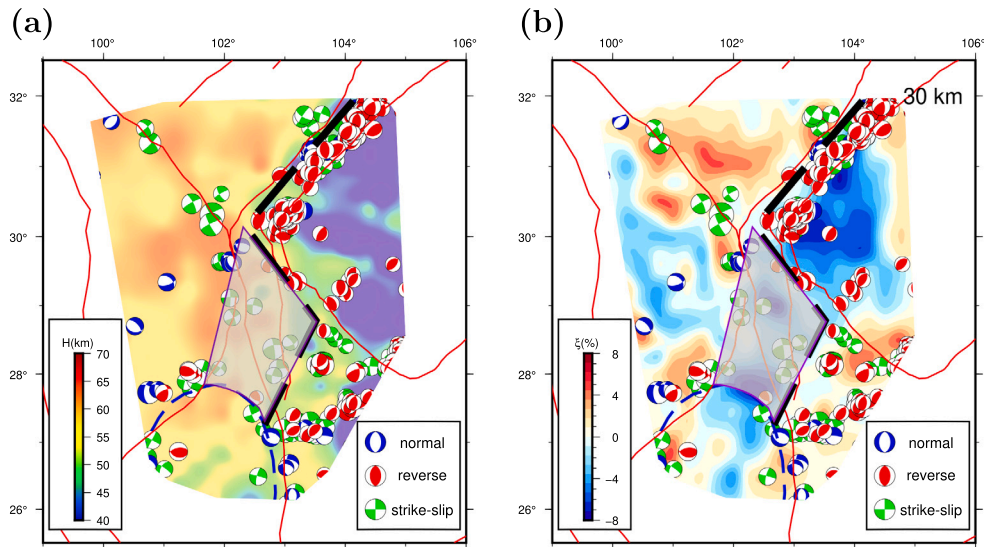


Fig. 5. Moho depths (a), radial anisotropy at depth of 30 km (b), and earthquake focal mechanisms in the study region. The Moho depths are according to Wang et al. (2017). The black dashed line divides the study region into two segments, east of which most focal mechanisms exhibit reverse faulting. The gray shaded area bounded by purple line outlines the area with thickened crust and negative radial anisotropy.

ies (Liu et al., 2020; Zheng et al., 2018; Zhang et al., 2022, 2023b; Cao et al., 2023), and our depth-dependent radial anisotropy model (Fig. 4).

5.2. The radial anisotropy related to earthquake focal mechanism and stress regime

The earthquake focal mechanisms used in this study are obtained from the China Seismic Experimental Site (<http://www.cses.ac.cn/sjcp/ggmx>). The focal mechanism catalog comprises 592 events with $M \geq 4$ recorded during 2009–2017, determined using the Cut-and-Paste method (Zhu and Helmberger, 1996).

To provide a comprehensive understanding of the tectonic stress regime, we classify the focal mechanisms into three groups: normal, reverse, and strike-slip faults, as shown in Fig. 5 (classification criteria can be found in Section 2 and Figure S1 in the supplementary material). A black dashed line along the eastern boundary of the Tibetan Plateau divides the region into two segments. East of this line, most focal mechanisms exhibit reverse faulting, which suggests that the area east of the Tibetan Plateau proper is predominantly under W-E to NW-SE compression. Notably, the northern portion of this line coincides with the LMSF, a well-known reverse fault, which ruptured during the 2008 Wenchuan earthquake (Burchfiel et al., 2008). Further south, the black dashed line delineates the boundaries of the SCB, the inner zone of the ELIP, the ANH-ZMHF, and the LJ-XJHF. This defines a region characterized by a thickened crust (Fig. 5a) and negative radial anisotropy (Fig. 5b) in the mid-lower crust, indicated by the gray shaded area bounded by the purple line in Fig. 5. This region is approximately situated along the boundary fault zone (BFZ) as designated by Cao et al. (2020).

Through the use of the P wave and S wave RF techniques, Hu et al. (2023) identified a mafic layer within the BFZ. Hu et al. (2023) and Peng et al. (2023) attributed this layer to intrusions and underplating of mafic material associated with the ancient Emeishan plume hotspot. This interpretation is supported by the high velocity and negative radial anisotropy observed in our models. Unlike the relatively stable SCB and the inner zone of the ELIP, the BFZ has likely undergone a complex geological evolution process. This complexity arises from the ANH-ZMHF traversing this region, effectively splitting the negative radial anisotropy into two sub-regions (Fig. 5b).

To the west of this area, most earthquakes are associated either with normal faulting, indicating extension of the upper crust, or strike slip. The latter mostly occur near known faults (like the left-lateral XSHF)

or in reverse fault zones with known strike-slip components (like the LMSF).

5.3. Implications for geodynamic evolution

In addition to the radial anisotropy described and interpreted above, one can infer azimuthal variations in shear wave speed – known as azimuthal anisotropy – from shear wave splitting. In Earth's crust, azimuthal anisotropy is often caused by the preferred orientation of minerals or geological fabric due to deformation. Comparing the polarization of core-refracted SKS splitting waves (Liu et al., 2020), and crustal Pms waves (Zheng et al., 2018), with GPS-derived strain rates (Wu et al., 2015), reveals that the directions of Pms splitting are intricate and align neither with the SKS splitting or with the orientations of GPS-derived strain rates (Figure S13). Along with the depth-dependent seismic azimuthal anisotropy found by Lev et al. (2006), Zhang et al. (2023b) and Cao et al. (2023), this misalignment suggests that the upper crust and mid-lower crust within the BFZ deform differently. This is consistent with the change from positive radial anisotropy at depths of 10–20 km to negative radial anisotropy above and below (Fig. 4f, h). Tan et al. (2019) proposed that the upper crust and mid-lower crust deform differently along the LMSF, with shortening and thrusting at shallow depths, and pure-shear thickening deeper in the crust. This interpretation could be applicable to the BFZ, with the additional complexity of the region and the regional strain potentially being accommodated by the strike-slip motion of the XSHF.

Another intriguing feature is the coincidence of weakly negative radial anisotropy in the western Chuan-Dian fragment at depths of 30–40 km (Fig. 3), and the predominance of earthquakes exhibiting normal faults (Fig. 5). Detailed analysis reveals that this area is characterized by E-W extensional tectonics, likely linked to the transition of strike-slip faults from their thrusting component to a normal component since the Middle-Late Miocene (Zhao et al., 2013; Tong et al., 2022). Tong et al. (2022) proposed that the subsidence and the extension of the uplifted Chuan-Dian fragment along preexisting strike-slip fault systems could have been the result of the reduction in the volume of lower crust flowing into the SETP due to the northward advance of the EHS since approximately 28 Ma. Independent thermochronological data imply that the SETP stopped expanding in the late Miocene (~10 Ma) (Shen et al., 2022). This geodynamical model could explain the weakly negative radial anisotropy observed in the western Chuan-Dian fragment because it suggests that horizontal flow is no longer prominent. It is worth noting,

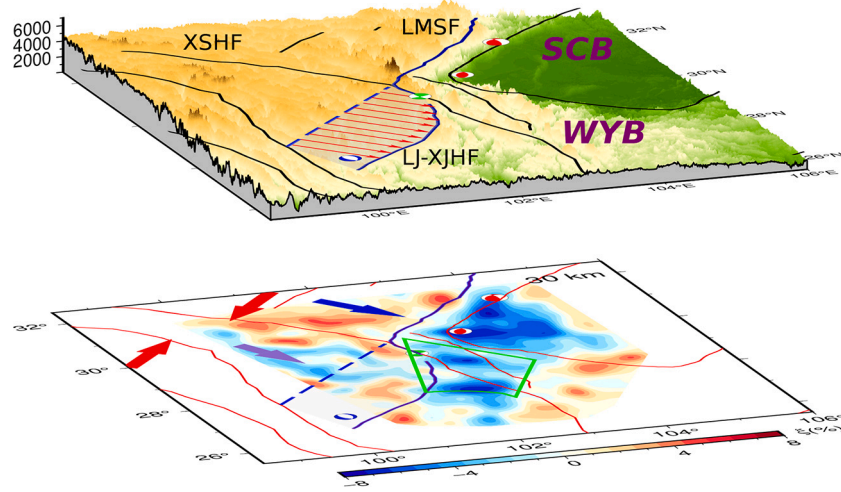


Fig. 6. Regional geodynamic evolution model. The red arrow represents the crustal motion caused by the northward advancing of EHS, resulting in the negative radial anisotropy in the western Chuan-Dian block. The purple arrow represents the southeastward crustal channel flow is impeded and the original topography can no longer be sustained, resulting the extension along LJ-XJHF. The blue arrow demonstrates the direction of the crustal movement, resulting in compression along LMSF. At the same time, the XSHF started to initiate and the mid-lower material with negative radial anisotropy within the BFZ were cut into pieces due to the faults activity (within green solid lines, details can be seen in Fig. 5). The red, green and blue beach balls indicate the recently occurred large earthquakes with reverse-, strike-slip, and normal-fault motion, respectively (details can be seen in Fig. 1).

however, that there may be alternative explanations for the negative radial anisotropy in the western Chuan-Dian fragment. Liu et al. (2014) proposed that “crustal channel flow” is not uniform and that the south-eastward expansion of crustal material may be influenced by various factors, including mechanical conditions, lateral variations in crustal structure and rheology, and strain partitioning across major faults. This non-uniform “crustal channel flow” may lead to the observed ambiguity of Moho-converted phases in RF waveforms (Liu et al., 2014) and potentially localized negative radial anisotropy in the mid-lower crust.

After taking into account the above considerations, we propose a regional geodynamic evolution with several stages. The key stages of the evolutionary process are briefly illustrated in Fig. 6 and detailed as follows:

- **Permian Plume Activity (Approximately 260 Ma):** Plume activity during the Permian period led to mafic intrusion and underplating in the inner, intermediate, and outer zones of the ELIP (Xu et al., 2006; Liu et al., 2021; Hu et al., 2023). The tectonic activity in this period may result in the high velocity zone in the inner zone around Panzhihua city, and the NE-SW directional belt zone in the BFZ with high velocity zone and negative radial anisotropy.
- **Northward Advance of the EHS (Since 28 Ma):** The EHS has been advancing northward since around 28 Ma, subjecting the Chuan-Dian fragment to a compressional stress field oriented NEE-SWW (Tong et al., 2022), and in the process changing the crustal fabric characterized by positive radial anisotropy (associated with horizontal flow) to weakly negative radial anisotropy.
- **Establishment of Modern Deformation Regime (Approximately 15 Ma):** By approximately 15 Ma, the modern deformation regime was established, characterized by N-S shortening in central Tibet and E-W extension in eastern Tibet (Royden et al., 2008). The continuous northward advance of the EHS impeded the southeastward flow of viscous lower crust and reduced the thickness of the lower crustal channel beneath the Chuan-Dian fragment. As a result, the original topography formed by the viscous lower crustal flow could no longer be sustained, leading to gravity-induced collapse and the development of an E-W extensional tectonic environment. This E-W extension may be linked to the occurrence of the 1996 Lijiang Mw 6.6 earthquake with a normal fault focal mechanism.
- **Major Tectonic Changes (Around 10 Ma):** Around the same time, at approximately 10 Ma, the heterogeneous distribution of strain

along the LMSF, LJ-XJHF, and other areas with different rigidity led to significant tectonic changes. The GS and the LMSF began to experience rapid uplift and exhumation, while the XSHF became active, collectively forming what could be referred to as the “huge eastern Tibet detachment system” (Zhang, 2013). Additionally, the LMSF-LJ-XJHF initiated their development, eventually taking on their current appearance (Gan et al., 2022).

- **Depth-dependent deformation and Strain Concentration (Present-Day):** In more recent times, the depth-dependent deformation exists in the upper crust and mid-lower crust in the BFZ. The strain concentration zone, which was originally near GS, appears to have moved southward and may now be situated near the ANHF (Yao and Yang, 2022). This southward shift in strain concentration could be associated with the Luding 2022 Mw 6.7 earthquake. Simultaneously, due to the stability of the SCB, the strain concentration zone has remained stationary. Consequently, the Wenchuan 2008 Mw 7.9 earthquake and the Lushan 2013 Mw 6.6 earthquake continue to occur along the LMSF.

This proposed geodynamic evolution model integrates various geological and geophysical observations to provide a comprehensive understanding of the tectonic history of the region. It highlights the complex interplay of geological processes and tectonic events that have shaped the present-day tectonic landscape.

By reconstructing the cooling and exhumation history of the Panxi tectonic belt using thermochronological data, Zhang et al. (2023a) proposed that there are two major tectonic deformation episodes since the late Eocene, that is, the periods of the late Eocene-Oligocene (36–26 Ma) and the Middle-Late Miocene (13–8 Ma). Their model aligns with ours, and these two models can mutually corroborate each other from different aspects.

6. Conclusions

In this study, we utilized continuous waveform data from a dense seismic array, combined with seismic ambient noise techniques, to simultaneously construct models for shear wave velocity (V_s) and regional radial anisotropy. These models offer a horizontal resolution of approximately 50 kilometers and were developed using the direct radial anisotropy tomography method. By assimilating these models with various geophysical and geological evidence, our findings suggest the

NE-SW directional belt-zone with negative radial anisotropy is due to the existence of mafic intrusion and underplating. Our results also imply the possibility of depth-dependent deformation in the upper and mid-lower crust within the boundary fault zone. Additionally, our models reveal a prevalent negative radial anisotropy in the western Chuan-Dian fragment, which can be attributed to ancient compressional forces. Moreover, we propose a geodynamic evolution model that aids in the interpretation of regional strain distribution and the genesis of recent significant seismic events.

CRedit authorship contribution statement

Shaoqian Hu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology. **Huajian Yao:** Writing – review & editing, Writing – original draft, Validation, Supervision, Methodology, Investigation, Funding acquisition, Conceptualization. **Jikun Feng:** Writing – review & editing, Visualization. **Hui Huang:** Writing – review & editing, Formal analysis, Data curation. **Qiyuan Liu:** Writing – review & editing. **Robert D. van der Hilst:** Writing – review & editing.

Declaration of competing interest

The authors acknowledge that there are no conflicts of interest recorded.

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Appendix A. Supplementary material

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.epsl.2024.119122>.

Data availability

Data will be made available on request.

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